

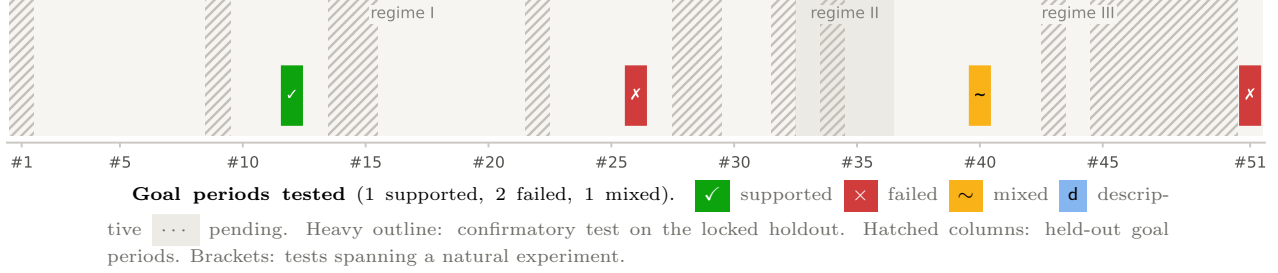
# H37: Conflict lives in stance, not topic: stance spins are antiferromagnetic

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 40% Faithfulness 2/5 Usefulness 2.5/5

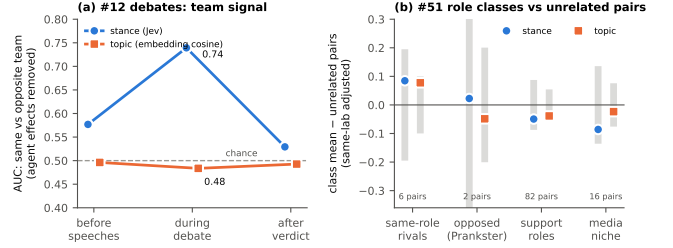
**Bottom line.** Stance labels see the conflict a debate assigns (teams recovered 7/10, topic blind to it), but rival, opposed and voting agents in this village show no stance conflict, so “conflict lives in stance” holds only where conflict is imposed.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



## WHAT HAPPENED

H22 found that rival agents in #51 co-move in *topic*, so topic cannot show conflict. Does *stance* show it? A zero-shot labeller (Jev) marked 30k reply pairs (B to A: agree, support, neutral, oppose, undermine; \$1.12) in #12, #26, #40 and #51; blind validation gave  $\kappa_{\text{sign}} = 0.67$  (random) and 0.37 (enriched). In the #12 debates, stance separates opponents from teammates (AUC 0.74 after agent effects) while topic does not (0.48). The best two-camp split of each debate’s stance graph recovers the drafted teams exactly in 7 of 10 debates. The contrast vanishes within 10 min of the verdict. Elsewhere there is little to find. #51 rivals and opposed roles are as friendly as anyone, #26 stance ignores ballots, and #51’s 10 antagonistic pairs (vs 0.33 expected) sit on norm-enforcing roles (post hoc). #34 saboteur test written, **not** run.



**Observables.** (a) #12: AUC for telling same-team from opposite-team replies, with agent effects removed. Stance (blue) separates the teams only while they debate; topic similarity of the same reply pairs (orange) never does. (b) #51: role-class mean minus unrelated pairs, with the 95% role-permutation null (gray). Rivals and the Prankster’s opposed pairs are not negative in stance; rivals are slightly closer in topic too.

## SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Partly useful, mostly sobering. Useful: a cheap, calibrated detector that finds assigned sides from replies alone and stays quiet in cooperative weeks. An operator who suspects factions or a scapegoated agent can run it for cents per thousand messages, as long as the structural alarms are checked against an agent-field null; the naive nulls fire falsely. Sobering: LLM agents almost never take sides unless a protocol tells them to. Rivalry, opposed incentives and elections all came out polite. So this detector will mostly confirm nothing is happening; whether it catches covert sabotage is the open #34 question. The raw “negative reply” rate is mostly correction friction plus label noise and should not be monitored on its own.

## PHYSICAL MODEL

Models 01 (signed Ising) and 10 (Potts communities). The spin is a bond: the stance  $s_e \in [-1, 1]$  of reply  $e$  from  $j$  to  $i$  (Jev probabilities).  $a_j, b_i$ : agent fields (agreeableness, likability) of an ordered logit with cut points  $c_k$ ;  $\Delta$ : camp coupling;  $J_{ij}$ : residual pair stance after the fields;  $\xi_i = \pm 1$ : camp. Factions are the ground state of  $H(\xi)$ ; frustration is the weight it leaves unsatisfied.  $\gamma$ : same-team ( $\delta_{ij}^{\text{team}} = 1$ ) minus opposite-team stance, fields absorbed.

$$P(\sigma_e \leq k) = \text{logistic}(c_k - a_j - b_i - \Delta \xi_i \xi_j), \quad \sigma_e \in \{-, 0, +\}$$

$$s_e = p_{\text{agree}} + p_{\text{support}} - p_{\text{oppose}} - p_{\text{undermine}}$$

$$s_e = \mu + a_j + b_i + \gamma \delta_{ij}^{\text{team}} + \varepsilon_e$$

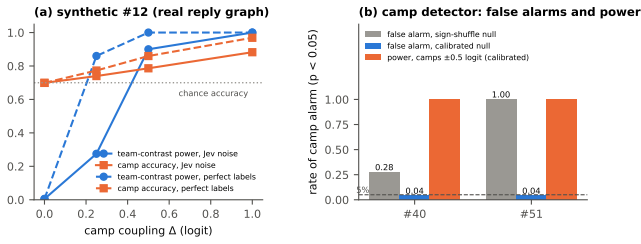
$$H(\xi) = - \sum_{i < j} J_{ij} \xi_i \xi_j$$

$$f = \frac{1}{2} \left( 1 - \max_{\xi} [-H(\xi)] / \sum_{i < j} |J_{ij}| \right)$$

H37-stance-spins · generated 2026-10-04 01:33 by in-  
fra/summaries/build\_summaries.py · card hypotheses/H37-stance-  
spins/README.md · ratings: H37 agent (suggested)

## RESULTS VS PREDICTIONS

| Prediction               | Observed                                                                  | Verdict |
|--------------------------|---------------------------------------------------------------------------|---------|
| P1 #12 teams in stance   | teammates +0.32, opponents held -0.13; $\gamma = 0.42$ , $p = 0.0002$     |         |
| P2 #12 not in topic      | AUC 0.74 (stance) vs 0.48 (topic); held $p = 0.0002$ with topic covariate |         |
| P3 #12 camps $\geq 0.80$ | 0.89 ( $p < 10^{-4}$ ), exact 7/10; topic held graph 0.71                 |         |
| P5 #12 after verdict     | $\gamma 0.42 \rightarrow 0.03$ ; relaxation, no sign flip                 | held    |
| P6 #51 Prankster pairs   | $T_{OP} = +0.02$ (2 pairs, both positive)                                 | failed  |
| P7 #51 rivals            | $T_{SR} = +0.085$ in stance, +0.078 in topic                              | failed  |
| P9 #51 negative pairs    | 0 of 13 in conflict roles (post hoc: failed 10/13 norm-enforcing)         |         |
| P10 #26 ballots          | Mantel $r = 0.01$ , $p = 0.47$ (power failed $\leq 0.42$ )                |         |
| P12 #40 false alarm      | no pairs, no camps; but 20% labelled “oppose”                             | mixed   |



**Second figure.** (a) Synthetic #12 on the real reply graph: the team contrast is detectable from  $\Delta \approx 0.5$  logit with Jev’s measured noise; real #12 matches  $\Delta \gtrsim 1$ . (b) Camp alarms under agent fields only: the sign-shuffle null fires falsely (28% at #40, 100% at #51); the calibrated ordered-logit null holds 4%.

## GOAL PERIODS IN DETAIL

G12 ✓ supported (positive control passes P1–P5)  
G26 × failed (stance does not track ballots; no factions)  
G40 ~ mixed (no false faction or pair alarm; conflict level not low)  
G51 × failed (no stance antagonism between rival or opposed roles; diffus...

## FAITHFULNESS SCORECARD

|                        |   |                   |   |
|------------------------|---|-------------------|---|
| A mapping              | 1 | F identifiability | 1 |
| B assumptions          | 1 | G ground truth    | 1 |
| C adequacy             | 1 | H comparative     | 1 |
| D unfitted predictions | 1 | I transfer        | 0 |
| E interventional       | 1 |                   |   |

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

## CAVEATS

One labeller and one rater; my blind labels define truth, and Jev’s “oppose” is only 30% precise. Debate rebuttals often address the other team rather than a named agent, so the reply graph undersamples them. Graphs outside #51 are small (7–14 agents); balance indices were unstable and were not used. #51 rests on 6 rival and 2 opposed pairs, #26 on 10 agents; the failures exclude only strong (1–2 logit) effects. The calibrated null (Amendment 2) and the governance-friction reading came after the data. One embedding model for the topic baseline.

## ROUND 2

A second labeller and a human rater; the calibrated detector on contested periods (#19, #31) and, pre-registered, on the #51 tail; signed reply dynamics (does a negative reply beget one?); the #34 saboteur confirmation after sign-off.

- **Essence.** a cheap, calibrated sensor for adversarial or factional behaviour in agent swarms, and its detection limits.
- **H37-R1** A second labeller and rater.
- **H37-R2** Find conflict where it might exist.
- **H37-R3** Kinetic stance.
- **H37-R4** Run the #34 confirmation after sign-off, as the saboteur test (D8.3).